



Status

Finished

Started

Tuesday, 9 June 2026, 9:30 AM

Completed

Tuesday, 9 June 2026, 12:07 PM

Duration

2 hours 37 mins

Question 1

Flag question

Complete

Marked out of 9.00

Machine assembly cost minimization.

A machine is assembled from components D_1 , D_2 , and D_3 that can be assembled using either bolts or rivets (Danish: bolte eller nitter).

Each machine requires 8 D_1 , 5 D_2 , and 15 D_3 components.

The assembly of D_1 requires either 5 bolts or 5 rivets, whereas D_2 requires 6 bolts or 6 rivets and D_3 requires 3 bolts or 3 rivets. The cost of installing 1 bolt, including the cost of the bolt itself, is 2.10 DKK for D_1 , 3.00 DKK for D_2 , and 1.80 DKK for D_3 . Similarly, riveting costs are 1.80 DKK for D_1 , 2.40 DKK for D_2 , and 3.00 DKK for D_3 .

This information is also stated in the following table:

	D_1 component	D_2 component	D_3 component
Number of units required per machine	8	5	15
Required bolts/rivets per component	5 bolts or 5 rivets	6 bolts or 6 rivets	3 bolts or 3 rivets
Cost of installing one bolt for the component	2.10 DKK	3.00 DKK	1.80 DKK
Cost of installing one rivet for the component	1.80 DKK	2.40 DKK	3.00 DKK

Bolting and riveting capacities per day are 6000 and 8000, respectively.

To minimize the cost for the 100 machines that must be assembled each day, we wish to determine the number of components to be bolted and riveted.

The minimum cost optimization problem should be formulated in standard form using Arora's five step procedure (step 1 and 2 are not needed).

Select the correct formulation (steps 3 to 5 and optimization problem).

Select one:

Step 3: Identify/definition of design variables. The following three design variables are identified:

x_1 : number of D_1 components assembled with bolts or rivets

x_2 : number of D_2 components assembled with bolts or rivets

x_3 : number of D_3 components assembled with bolts or rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost

objective f is identified:

$$f(\mathbf{x}) = (2.10 + 1.80)x_1 + (3.00 + 2.40)x_2 + (1.80 + 3.00)x_3$$

Step 5: Identification of constraints.

Component requirements per machine:

$$x_1 = 8 \cdot 100 = 800$$

$$x_2 = 5 \cdot 100 = 500$$

☐ a. $x_3 = 15 \cdot 100 = 1500$

Bolting and riveting capacities:

$$5x_1 + 6x_2 + 3x_3 \leq 6000 + 8000$$

Non-negativity constraints:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = (2.10 + 1.80)x_1 + (3.00 + 2.40)x_2 + (1.80 + 3.00)x_3$$

Subject to:

$$x_1 = 800$$

$$x_2 = 500$$

$$x_3 = 1500$$

$$5x_1 + 6x_2 + 3x_3 \leq 14000$$

$$-x_1 \leq 0, -x_2 \leq 0, -x_3 \leq 0$$

Step 3: Identify/definition of design variables. The following six design variables are identified:

x_1 : number of D_1 components assembled with bolts

x_2 : number of D_1 components assembled with rivets

x_3 : number of D_2 components assembled with bolts

x_4 : number of D_2 components assembled with rivets

x_5 : number of D_3 components assembled with bolts

x_6 : number of D_3 components assembled with rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost objective f is identified:

$$f(\mathbf{x}) = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

Step 5: Identification of constraints.

Component requirements per machine:

$$x_1 + x_2 \leq 8 \cdot 100 = 800$$

☐ b. $x_3 + x_4 \leq 5 \cdot 100 = 500$

$$x_5 + x_6 \leq 15 \cdot 100 = 1500$$

Bolting and riveting capacities:

$$5x_1 + 6x_3 + 3x_5 \leq 6000$$

$$5x_2 + 6x_4 + 3x_6 \leq 8000$$

Non-negativity constraints:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

Subject to:

$$x_1 + x_2 \leq 800$$

$$x_3 + x_4 \leq 500$$

$$x_5 + x_6 \leq 1500$$

$$5x_1 + 6x_3 + 3x_5 \leq 6000$$

$$5x_2 + 6x_4 + 3x_6 \leq 8000$$

$$-x_1 \leq 0, -x_2 \leq 0, -x_3 \leq 0, -x_4 \leq 0, -x_5 \leq 0, -x_6 \leq 0$$

- ☐ c. None of the other solutions

Step 3: Identify/definition of design variables. The following six design variables are identified:

x_1 : number of D_1 components assembled with bolts

x_2 : number of D_1 components assembled with rivets

x_3 : number of D_2 components assembled with bolts

x_4 : number of D_2 components assembled with rivets

x_5 : number of D_3 components assembled with bolts

x_6 : number of D_3 components assembled with rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost objective f is identified:

$$f(\mathbf{x}) = 2.10x_1 + 1.80x_2 + 3.00x_3 + 2.40x_4 + 1.80x_5 + 3.00x_6$$

Step 5: Identification of constraints.

Component requirements per machine:

$$x_1 + x_2 = 8 \cdot 100 = 800$$

$$x_3 + x_4 = 5 \cdot 100 = 500$$

- ☐ d. $x_5 + x_6 = 15 \cdot 100 = 1500$

Bolting and riveting capacities:

$$5x_1 + 6x_3 + 3x_5 \leq 6000$$

$$5x_2 + 6x_4 + 3x_6 \leq 8000$$

Non-negativity constraints:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = 2.10x_1 + 1.80x_2 + 3.00x_3 + 2.40x_4 + 1.80x_5 + 3.00x_6$$

Subject to:

$$x_1 + x_2 = 800$$

$$x_3 + x_4 = 500$$

$$x_5 + x_6 = 1500$$

$$5x_1 + 6x_3 + 3x_5 \leq 6000$$

$$5x_2 + 6x_4 + 3x_6 \leq 8000$$

$$-x_1 \leq 0, \quad -x_2 \leq 0, \quad -x_3 \leq 0, \quad -x_4 \leq 0, \quad -x_5 \leq 0, \quad -x_6 \leq 0$$

Step 3: Identify/definition of design variables. The following six design variables are identified:

x_1 : number of D_1 components assembled with bolts

x_2 : number of D_1 components assembled with rivets

x_3 : number of D_2 components assembled with bolts

x_4 : number of D_2 components assembled with rivets

x_5 : number of D_3 components assembled with bolts

x_6 : number of D_3 components assembled with rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost objective f is identified:

$$f(\mathbf{x}) = 5(2.10x_1 + 1.80x_2) + 6(3.00x_3 + 2.40x_4) + 3(1.80x_5 + 3.00x_6)$$

Step 5: Identification of constraints.

Component requirements per machine:

$$x_1 + x_2 = 8 \cdot 100 = 800$$

$$x_3 + x_4 = 5 \cdot 100 = 500$$

e. $x_5 + x_6 = 15 \cdot 100 = 1500$

Bolting and riveting capacities:

$$5x_1 + 6x_3 + 3x_5 \leq 6000$$

$$5x_2 + 6x_4 + 3x_6 \leq 8000$$

Non-negativity constraints:

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0, \quad x_4 \geq 0, \quad x_5 \geq 0, \quad x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = 5(2.10x_1 + 1.80x_2) + 6(3.00x_3 + 2.40x_4) + 3(1.80x_5 + 3.00x_6)$$

Subject to:

$$x_1 + x_2 = 800$$

$$x_3 + x_4 = 500$$

$$x_5 + x_6 = 1500$$

$$5x_1 + 6x_3 + 3x_5 \leq 6000$$

$$5x_2 + 6x_4 + 3x_6 \leq 8000$$

$$-x_1 \leq 0, \quad -x_2 \leq 0, \quad -x_3 \leq 0, \quad -x_4 \leq 0, \quad -x_5 \leq 0, \quad -x_6 \leq 0$$

Step 3: Identify/definition of design variables. The following six design variables are identified:

x_1 : number of D_1 components assembled with bolts

x_2 : number of D_1 components assembled with rivets

x_3 : number of D_2 components assembled with bolts

x_4 : number of D_2 components assembled with rivets

x_5 : number of D_3 components assembled with bolts

x_6 : number of D_3 components assembled with rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost objective f is identified:

$$f(\mathbf{x}) = 5(x_1 + x_2) + 6(x_3 + x_4) + 3(x_5 + x_6)$$

Step 5: Identification of constraints.

Price requirements per machine:

$$2.10x_1 + 1.80x_2 \leq 8 \cdot 100 = 800$$

☐ f. $3.00x_3 + 2.40x_4 \leq 5 \cdot 100 = 500$

$$1.80x_5 + 3.00x_6 \leq 15 \cdot 100 = 1500$$

Bolting and riveting capacities:

$$x_1 + x_3 + x_5 \leq 6000$$

$$x_2 + x_4 + x_6 \leq 8000$$

Non-negativity constraints:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = 5(2.10x_1 + 1.80x_2) + 6(3.00x_3 + 2.40x_4) + 3(1.80x_5 + 3.00x_6)$$

Subject to:

$$2.10x_1 + 1.80x_2 \leq 800$$

$$3.00x_3 + 2.40x_4 \leq 500$$

$$1.80x_5 + 3.00x_6 \leq 1500$$

$$x_1 + x_3 + x_5 \leq 6000$$

$$x_2 + x_4 + x_6 \leq 8000$$

$$-x_1 \leq 0, -x_2 \leq 0, -x_3 \leq 0, -x_4 \leq 0, -x_5 \leq 0, -x_6 \leq 0$$

Step 3: Identify/definition of design variables. The following three design variables are identified:

x_1 : number of D_1 components assembled with bolts or rivets

x_2 : number of D_2 components assembled with bolts or rivets

x_3 : number of D_3 components assembled with bolts or rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost objective f is identified:

$$f(\mathbf{x}) = 5(2.10 + 1.80)x_1 + 6(3.00 + 2.40)x_2 + 3(1.80 + 3.00)x_3$$

Step 5: Identification of constraints.

Component requirements per machine:

$$x_1 = 8 \cdot 100 = 800$$

$$x_2 = 5 \cdot 100 = 500$$

☐ g. $x_3 = 15 \cdot 100 = 1500$

Bolting and riveting capacities:

$$5x_1 + 6x_2 + 3x_3 \leq 6000 + 8000$$

Non-negativity constraints:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = 5(2.10 + 1.80)x_1 + 6(3.00 + 2.40)x_2 + 3(1.80 + 3.00)x_3$$

Subject to:

$$x_1 = 800$$

$$x_2 = 500$$

$$x_3 = 1500$$

$$5x_1 + 6x_2 + 3x_3 \leq 14000$$

$$-x_1 \leq 0, -x_2 \leq 0, -x_3 \leq 0$$

Step 3: Identify/definition of design variables. The following six design variables are identified:

x_1 : number of D_1 components assembled with bolts

x_2 : number of D_1 components assembled with rivets

x_3 : number of D_2 components assembled with bolts

x_4 : number of D_2 components assembled with rivets

x_5 : number of D_3 components assembled with bolts

x_6 : number of D_3 components assembled with rivets

Step 4: Identification of a criterion to be optimized. The following minimum cost objective f is identified:

$$f(\mathbf{x}) = 2.10x_1 + 1.80x_2 + 3.00x_3 + 2.40x_4 + 1.80x_5 + 3.00x_6$$

Step 5: Identification of constraints.

Component requirements per machine:

$$x_1 + x_2 = 8$$

$$x_3 + x_4 = 5$$

☐ h. $x_5 + x_6 = 15$

Bolting and riveting capacities:

$$x_1 + x_3 + x_5 \leq 6000$$

$$x_2 + x_4 + x_6 \leq 8000$$

Non-negativity constraints:

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0, x_6 \geq 0$$

The optimization problem in standard form becomes:

$$\text{minimize } f(\mathbf{x}) = 2.10x_1 + 1.80x_2 + 3.00x_3 + 2.40x_4 + 1.80x_5 + 3.00x_6$$

Subject to:

$$x_1 + x_2 = 8$$

$$x_3 + x_4 = 5$$

$$x_5 + x_6 = 15$$

$$x_1 + x_3 + x_5 \leq 6000$$

$$x_2 + x_4 + x_6 \leq 8000$$

$$-x_1 \leq 0, -x_2 \leq 0, -x_3 \leq 0, -x_4 \leq 0, -x_5 \leq 0, -x_6 \leq 0,$$

Question 2

[Flag question](#)

Complete

Marked out of 8.00

The following constrained minimisation problem is considered:

$$\text{minimize } f(\mathbf{x}) = (2 + x_1)^2 + (3 + x_2)^2$$

Subject to the inequality constraints:

$$g_1(\mathbf{x}) = x_1 + x_2 + 3 \leq 0$$

$$g_2(\mathbf{x}) = -x_1 + x_2 + 3 \leq 0$$

Determine the optimum solution to the problem using graphical methods and/or the KKT necessary conditions.

Select one:

- ☐ a. $x_1 = 1, x_2 = -4$
- ☐ b. $x_1 = -1, x_2 = 4$
- ☐ c. None of the other solutions
- ☐ d. $x_1 = 1, x_2 = 4$
- ☐ e. $x_1 = 4, x_2 = 1$
- ☐ f. $x_1 = 0, x_2 = -3$
- ☐ g. $x_1 = 0, x_2 = 3$
- ☒ h. $x_1 = -1, x_2 = -4$
- ☐ i. $x_1 = 1, x_2 = 3$

☐ J. $x_1 = -4, x_2 = -1$

Question 3

[Flag question](#)

Complete

Marked out of 3.00

We will consider gradient-based minimisation of the following unconstrained function:

$$f(\mathbf{x}) = (1 - x_1)^2 + (x_2 - 2)^2 + x_1 x_2$$

The starting point is: $\mathbf{x}^{(0)} = [2, -1]^T$.

In order to complete the first iteration of the modified Newton's method for the function, the Hessian must be evaluated. What is the expression for the Hessian, $\mathbf{H}(\mathbf{x})$?

Select one:

☐ a. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

☐ b. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$

☐ c. None of the other solutions

☐ d. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 8 & -4 \\ -4 & 2 \end{bmatrix}$

☒ e. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

☐ f. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

☐ g. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$

☐ h. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}$

☐ i. $\mathbf{H}(\mathbf{x}) = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

Question 4

[Flag question](#)

Complete

Marked out of 3.00

In order to complete the first iteration of the modified Newton's method for the function above using the analytical approach, the 1D line search problem should be established.

What is the expression for $f(\alpha)$, where α is the step size in direction $\mathbf{d}^{(0)}$?

Select one:

- ☐ a. None of the other solutions
- ☐ b. $f(\alpha) = \alpha^2 - 2\alpha + 2$
- ☐ c. $f(\alpha) = -11\alpha^2 + 22\alpha - 10$
- ☐ d. $f(\alpha) = (3\alpha^2 - 6\alpha + 4)$
- ☐ e. $f(\alpha) = (61\alpha^2)/3 - (122\alpha)/3 + 16$
- ☐ f. Newton's method cannot be applied as the Hessian is not positive definite
- ☒ g. $7\alpha^2 - 14\alpha + 8$
- ☐ h. $f(\alpha) = (13\alpha^2)/3 - 26\alpha/3$
- ☐ i. $f(\alpha) = 7\alpha^2 + 14\alpha + 8$

Question 5

[Flag question](#)

Complete

Marked out of 2.00

Complete the first iteration of the modified Newton's method for the function above.

What are the values of the design variables after the first iteration, $\mathbf{x}^{(1)}$?

Select one:

☐ a. $\mathbf{x}^{(1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

☐ b. $\mathbf{x}^{(1)} = \begin{bmatrix} 8/5 \\ 6/5 \end{bmatrix}$

☐ c. None of the other solutions

☐ d. $\mathbf{x}^{(1)} = \begin{bmatrix} -8/3 \\ 10/3 \end{bmatrix}$

☐ e. $\mathbf{x}^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

☐ f. $\mathbf{x}^{(1)} = \begin{bmatrix} 8/3 \\ -10/3 \end{bmatrix}$

☐ g. $\mathbf{x}^{(1)} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

☒ h. $\mathbf{x}^{(1)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

☐ i. Newton's method cannot be applied as the Hessian is not positive definite

Question 6

Flag question

Complete

Marked out of 3.00

An optimisation problem is formulated as:

$$\text{minimize } f(x) = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2$$

Subject to:

$$\begin{aligned} g_1(\mathbf{x}) &= -x_1 + x_2 - 2 \leq 0 \\ g_2(\mathbf{x}) &= \frac{(x_1-1)^2}{9} + \frac{(x_2-1)^2}{4} - 1 \leq 0 \end{aligned}$$

The problem is to be solved using the alternate form of the Karush-Kuhn-Tucker conditions.

Determine the Lagrangian for the problem:

Select one:

- ☐ $L = -(x_1 - 5)^2 - \frac{1}{4}(x_2 - 5)^2$
- ☐ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2$
- ☐ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2 + u_1 (-x_1 + x_2 - 2 + s_1^2)$
- ☐ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2 + u_2 \left(\frac{(x_1-1)^2}{9} + \frac{(x_2-1)^2}{4} - 1 + s_2^2 \right)$
- ☐ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2 + u_2 \left(\frac{(x_1-1)^2}{9} + \frac{(x_2-1)^2}{4} - 1 \right)$
- ☐ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2 + u_1 (-x_1 + x_2 - 2)$
- ☒ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2 + u_1 (-x_1 + x_2 - 2) + u_2 \left(\frac{(x_1-1)^2}{9} + \frac{(x_2-1)^2}{4} - 1 \right)$
- ☐ $L = (x_1 - 5)^2 + \frac{1}{4}(x_2 - 5)^2 + u_1 (-x_1 + x_2 - 2 + s_1^2) + u_2 \left(\frac{(x_1-1)^2}{9} + \frac{(x_2-1)^2}{4} - 1 + s_2^2 \right)$
- ☐ Value not described by any of the other answer options

Question 7

[Flag question](#)

Complete

Marked out of 2.00

Correct answers yield positive points, while wrong answers count negatively. The total score of this sub-question cannot be negative.

Determine the gradient conditions for the problem:

Select one or more:



$$\frac{\partial L}{\partial x_2} = \frac{x_2 - 5}{2} + u_1 + \frac{u_2}{2}(x_2 - 1) = 0$$



$$\frac{\partial L}{\partial x_2} = \frac{x_2 - 5}{2} + u_1 = 0$$



$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 - s_1^2 + s_2^2 = 0$$



$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 - u_1 s_1^2 + \frac{u_2 s_2^2}{9}(2x_1 - 2) = 0$$



$$\frac{\partial L}{\partial x_2} = \frac{x_2 - 5}{2} + s_1^2 + s_2^2 = 0$$



$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 + \frac{u_2}{9}(2x_1 - 2) = 0$$



$$\frac{\partial L}{\partial x_2} = \frac{x_2 - 5}{2} = 0$$



$$\frac{\partial L}{\partial x_2} = \frac{x_2 - 5}{2} + \frac{u_2}{2}(x_2 - 1) = 0$$



$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 - u_1 + \frac{u_2}{9}(2x_1 - 2) = 0$$



Expression not described by any of the other answer options



$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 - u_1 = 0$$

☐
$$\frac{\partial L}{\partial x_2} = \frac{x_2 - 5}{2} + u_1 s_1^2 + \frac{u_2 s_2^2}{2} (x_2 - 1) = 0$$

☐
$$\frac{\partial L}{\partial x_1} = 2x_1 - 10 = 0$$

Question 8

[Flag question](#)

Complete

Marked out of 2.00

Correct answers yield positive points, while wrong answers count negatively. The total score of this sub-question cannot be negative.

Determine the switching conditions for the problem:

Select one or more:

☐ Expression not described by any of the other answer options

☐
$$u_2 \left(\frac{(x_1 - 1)^2}{9} + \frac{(x_2 - 1)^2}{4} - 1 + x_2^2 \right) = 0$$

☒
$$u_1 (-x_1 + x_2 - 2) = 0$$

☐
$$u_1 (x_1 - 5)^2 = 0$$

☒
$$u_2 \left(\frac{(x_1 - 1)^2}{9} + \frac{(x_2 - 1)^2}{4} - 1 \right) = 0$$

☐
$$u_2 \left(\frac{1}{4} (x_2 - 5)^2 \right) = 0$$

☐
$$u_1 (-x_1 + x_2 - 2 + s_1^2) = 0$$

Question 9

[Flag question](#)

Complete

Marked out of 4.00

Correct answers yield positive points, while wrong answers count negatively. The total score of this sub-question cannot be negative.

Determine the coordinates of the candidate point(s) (x_1^*, x_2^*) and the associated Lagrange multipliers (u_1^*, u_2^*) fulfilling the necessary conditions:

Select one or more:

- ☐ $x_1^* = 5$
- ☐ $u_2^* = 0.09$
- ☐ $u_1^* = 8.03$
- ☐ $x_2^* = 1$
- ☐ $u_2^* = 0$
- ☐ $u_2^* = 1$
- ☐ $x_1^* = 1.72$
- ☐ $x_2^* = 0.87$
- ☐ $x_2^* = 2.71$
- ☒ $u_2^* \approx 4.06$
- ☒ $x_2^* \approx 1.79$

- ☐ $x_2^* = 5$
- ☐ $x_1^* = 3.09$
- ☐ $x_1^* = 1$
- ☐ $u_1^* = 1$
- ☒ $u_1^* = 0$
- ☐ Expression not described by any of the other answer options
- ☐ $u_1^* = 3.67$
- ☐ $u_2^* = 1.82$
- ☐ $u_1^* = 0.26$
- ☒ $x_1^* \approx 3.76$

Question 10

[Flag question](#)

Complete

Marked out of 1.00

Determine the Hessian matrix:

Select one:

☐ $H \approx \begin{bmatrix} 5.41 & 0 \\ 0 & 1.65 \end{bmatrix}$

☐ $H \approx \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

☒ $H \approx \begin{bmatrix} 2.90 & 0 \\ 0 & 2.53 \end{bmatrix}$

☐ $H \approx \begin{bmatrix} 5.41 & 3.22 \\ 1.72 & 1.65 \end{bmatrix}$

☐ $H \approx \begin{bmatrix} 2.41 & -1.12 \\ -0.49 & 5.14 \end{bmatrix}$

☐ $H \approx \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

☐ $H \approx \begin{bmatrix} 3.71 & 2.45 \\ 0 & 9.84 \end{bmatrix}$

☐ $H \approx \begin{bmatrix} 0.21 & 5.41 \\ 0.23 & 8.41 \end{bmatrix}$

☐ Expression not described by any of the other answer options

Question 11

[Remove flag](#)

Complete

Marked out of 1.00

Determine which of the following statements are true for the candidate point:

Select one:

- ☐ The candidate point is an isolated local maximum point
- ☐ The candidate point is not characterized by any of the other answer options
- ☐ The candidate point is an inflection point
- ☐ The candidate point is a local maximum point
- ☐ The candidate point is an isolated minimum point
- ☐ The candidate point is an isolated maximum point
- ☒ The candidate point is not an optimum point
- ☐ The candidate point is an isolated local minimum point
- ☐ The candidate point is a local minimum point

Question 12[Flag question](#)

Complete

Marked out of 4.00

Correct answers yield positive points, while wrong answers count negatively. The total score of this sub-question cannot be negative.

In a given multi-objective optimization problem the objective functions are implicit (unknown), as the function values are output from a simulation program. For this reason a number of designs have been generated, and the corresponding objective function values for each design are given in the below table, which has been sorted according to increasing function value for the first design objective (f_1).

Design: (f_1, f_2)	Design: (f_1, f_2)	Design: (f_1, f_2)
D1: $(-6.0, 56.3)$	D11: $(1.1, 34.2)$	D21: $(33.3, 4.0)$
D2: $(-5.5, 49.4)$	D12: $(7.0, 22.1)$	D22: $(35.6, 6.9)$
D3: $(-4.2, 42.9)$	D13: $(8.2, 23.0)$	D23: $(49.3, 0.8)$
D4: $(-4.0, 52.3)$	D14: $(17.1, 11.7)$	D24: $(49.8, 2.5)$
D5: $(-3.8, 58.3)$	D15: $(17.8, 12.7)$	D25: $(52.6, 2.6)$
D6: $(-3.6, 52.8)$	D16: $(18.6, 12.6)$	D26: $(53.5, 1.9)$
D7: $(-3.5, 47.0)$	D17: $(20.6, 12.7)$	D27: $(75.1, -0.8)$
D8: $(-2.6, 53.3)$	D18: $(23.0, 7.9)$	D28: $(78.2, 1.0)$
D9: $(-0.5, 33.6)$	D19: $(24.7, 9.5)$	D29: $(79.4, 2.1)$
D10: $(1.0, 35.0)$	D20: $(26.0, 10.8)$	D30: $(80.1, -0.3)$

Determine which of the design points that are Pareto optimal (assume the non-dominated points to be Pareto optimal).

Select one or more:

D1: $(-6.0, 56.3)$ D2: $(-5.5, 49.4)$

— , , ,

D3: $(-4.2, 42.9)$



☐ D4: $(-4.0, 52.3)$

☐ D5: $(-3.8, 58.3)$

☐ D6: $(-3.6, 52.8)$

☐ D7: $(-3.5, 47.0)$

☐ D8: $(-2.6, 53.3)$

☒ D9: $(-0.5, 33.6)$

☐ D10: $(1.0, 35.0)$

☐ D11: $(1.1, 34.2)$

☒ D12: $(7.0, 22.1)$

☐ D13: $(8.2, 23.0)$

☒ D 14: $(17.1, 11.7)$

☐ D15: $(17.8, 12.7)$

☐ D16: $(18.6, 12.6)$

☐ D 17: $(20.6, 12.7)$

☒ D18: $(23.0, 7.9)$

☐ D19: $(24.7, 9.5)$

☐

D20: (26.0, 10.8)

☒

D21: (33.3, 4.0)

☐

D22: (35.6, 6.9)

☒

D23: (49.3, 0.8)

☐

D24: (49.8, 2.5)

☐

D25: (52.6, 2.6)

☐

D26: (53.5, 1.9)

☒

D27: (75.1, -0.8)

☐

D28: (78.2, 1.0)

☐

D29: (79.4, 2.1)

☐

D30: (80.1, -0.3)

☐

None of the points are Pareto Optimal

Question 13

[Flag question](#)

Complete

Marked out of 2.00

Based on the above points determine the utopia point:

Select one:

- ☒ $(-6.0, -0.8)$
- ☐ $(-8.1, -2.0)$
- ☐ $(1.0, -0.8)$
- ☐ $(-4.2, 1.0)$
- ☐ $(-0.3, -4.2)$
- ☐ $(-3.8, -4.2)$
- ☐ The coordinates for the utopia point is not include in the other answer options
- ☐ $(-5.5, -0.3)$

Question 14

[Flag question](#)

Complete

Marked out of 2.00

Determine which of the following points in the solution space that is the closest to the compromise solution:

Select one:

☒

D14

☐

None of the points are closest to the compromise solution

☐

D15

☐

D12

☐

D16

☐

D18

Question 15

[Remove flag](#)

Complete

Marked out of 4.00

Correct answers yield positive points, while wrong answers count negatively. The total score of this sub-question cannot be negative.

Mark each of the statements that are correct:

Select one or more:

- ☐ Exact line search is generally preferred in practical nonlinear optimization routines because it provides the fastest convergence.
- ☐ Genetic algorithms do not rely on gradient information and can be broadly applied.
- ☒ From the reduced cost coefficients in the simplex tableau, it is possible to determine if an optimal solution has been reached.
- ☐ Using a weighted sum method will generate all Pareto optimal solutions for any multi-objective optimisation problem.
- ☐ Selection in genetic algorithms always guarantees that the best solution survives to the next generation.
- ☐ SQP-based approaches solve a quadratic programming subproblem at each iteration to approximate the nonlinear problem.
- ☒ Mutations and permutations in genetic algorithms are included to prevent premature convergence.
- ☐ In SQP-problems both the objective function and the constraints are approximated by quadratic functions.
- ☐ Every feasible linear optimization problem has an optimal solution.
- ☒ It is a requirement in a standard linear programming problem that both the objective function and constraints are linear.

☐

The linear Simplex method is based on evaluating the gradient of the objective function.



For a convex problem it is always guaranteed to find the global optimum with a numerically based optimisation approach.

Question 16

[Remove flag](#)

Complete

Marked out of 3.00

Please select all the correct descriptions from the following about different types of reliability testing:

Select one or more:

- ☒ a. For ALT (Accelerated Life Test) aiming to test-to-failure of most of the samples, 6 to 12 sample size is a proper trade-off between the statistical uncertainty and required number of samples.
- ☒ b. HALT (Highly Accelerated Life Test) is an effective testing method to identify the weakest points of the testing items.
- ☐ c. HALT (Highly Accelerated Life Test) aims to obtain the lifetime model of the testing items under specific stress conditions, such as temperature, humidity, and vibration.
- ☒ d. Typical CALT (Calibrated Accelerated Life Test) determine the stress level of the 3rd group of testing according to the available testing time.
- ☐ e. RSS (Reliability Stress Screening) is usually implemented with accelerated stress conditions beyond product specifications in order to reduce the testing time.

Question 17

[Flag question](#)

Complete

Marked out of 2.00

It requires to demonstrate that an electronic component has reliability $R = 0.99$ at confidence level $CL = 90\%$ at the end of one equivalent life of 1000 thermal cycles. Success-run test strategy is implemented.

The required sample size for 1000 thermal cycle testing is (round up to the next integer):

Answer:

Question 18

[Flag question](#)

Complete

Marked out of 3.00

In connection with the previous question, the reliability of the electronic component related to thermal-mechanical fatigue follows Weibull distribution with the shaping factor $\beta = 3$. If there are only 20 samples available and the testing condition is kept the same, the minimum number of thermal cycles to test is (round up to the next integer):

Answer:

Question 19

[Flag question](#)

Complete

Marked out of 2.00

If $\beta = 1.5$ for a given type of battery used in laptops. It means the hazard rate of that type of battery is

Select one:

- ☒ a. Increasing with time
- ☐ b. Constant with time
- ☐ c. With unknown trend
- ☐ d. Decreasing with time

Question 20

[Flag question](#)

Complete

Marked out of 2.00

Two types of batteries are the candidates for a laptop design. The time-to-failure data of both Type A and Type B batteries can fit well by a two-parameter Weibull distribution. With the same operation condition, same end-of-life criteria, and a confidence level of 50%, if Type A battery has a B1 lifetime of 800 full charging/discharging cycles and Type B battery has a B10 lifetime of 1200 full charging/discharging cycles, which type of battery has a higher reliability level at the designed service life of 600 full charging/discharging cycles?

Select one:

- ☒ a. Unknown
- ☐ b. Type A and Type B have equal reliability level
- ☐ c. Type B
- ☐ d. Type A

Question 21

[Flag question](#)

Complete

Marked out of 2.00

What is the corresponding confidence level of a Weibull plot when the Median rank of the time-to-failure data is applied?

Select one:

- ☒ a. 50%
- ☐ b. 63.2%
- ☐ c. 36.8%
- ☐ d. Unknown

Question 22

[Flag question](#)

Complete

Marked out of 3.00

In an accelerated life test of a type of rechargeable battery. 9 battery samples are used. At the end of the test, one battery sample still survives. Median rank is used to find the corresponding unreliability level of the testing samples for Weibull analysis. What are the corresponding unreliability levels for the first failed sample and the last failed sample?

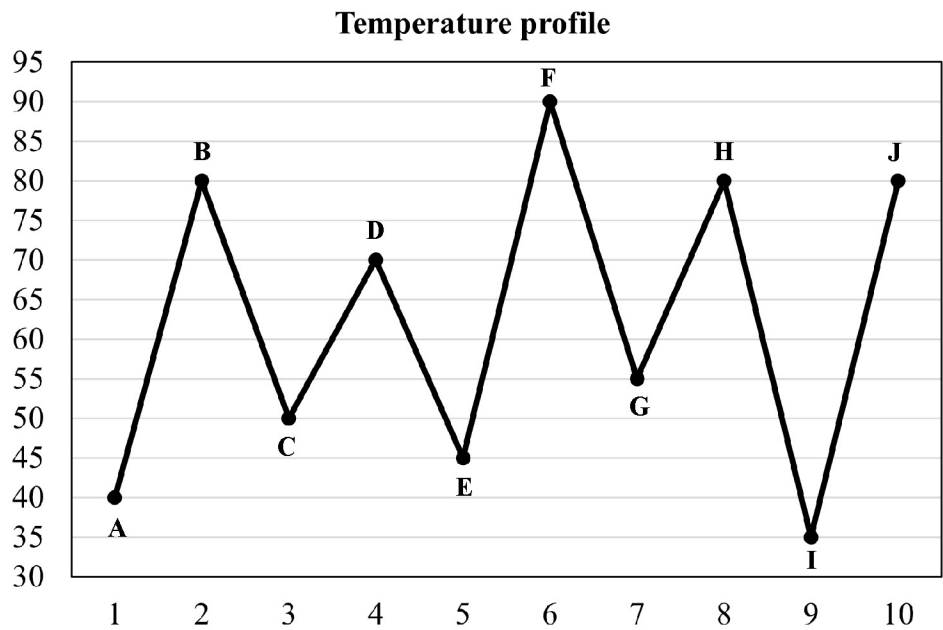
- ☒ a. 0.0745 and 0.8191
- ☐ b. 0.1000 and 0.8000
- ☐ c. 0.0745 and 0.9255
- ☐ d. 0.0833 and 0.9167

Question 23

Flag question

Complete Marked out of 4.00

An electronic component operates with the following temperature stress profile. The data points are sampled every one second. Which of the following table shows the correct Rainflow counting results?
40, 80, 50, 70, 45, 90, 55, 80, 35, 80 [°C]



Select one:



a.

Cycle range	Cycle counts	Cycle events
20	1	C-D
25	1	G-H
35	1	B-E
45	1	H-I, I-J
50	0.5	A-F



b.

Cycle range	Cycle counts	Cycle events
20	0.5	C-D
25	1	D-E, G-H
30	0.5	B-C
40	0.5	A-B
45	1.5	E-F, H-I, I-J

35	0.5	F-G
----	-----	-----



c.

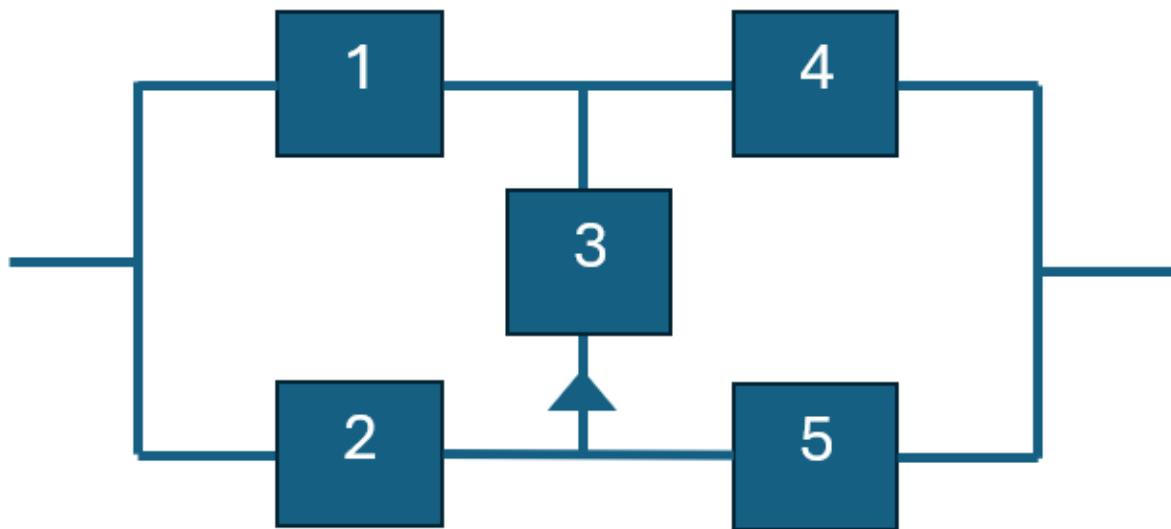
Cycle range	Cycle counts	Cycle events
20	1	C-D
25	1	G-H
35	1	B-E
45	0.5	I-J
50	0.5	A-F
55	0.5	F-I

Question 24[Flag question](#)

Complete

Marked out of 3.00

We have a unidirectional bridge shown in this figure. Component 3 works only in one direction to pass the input to the output. What are the minimal cut sets of this system?



Select one or more:

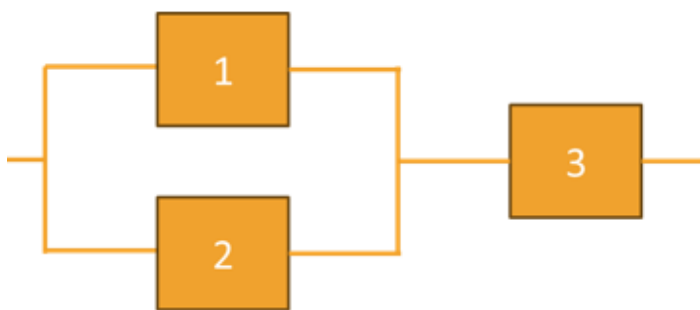
- ☐ a. {2, 4}
- ☐ b. {1, 3, 5}
- ☒ c. {4, 5}
- ☐ d. {1, 5}
- ☒ e. {1, 2}
- ☒ f. {2, 3, 4}

Question 25[Flag question](#)

Complete

Marked out of 3.00

We have a system including 3 components as following figure. The reliability of each component is 0.7. What is the reliability of entire system?



Select one:

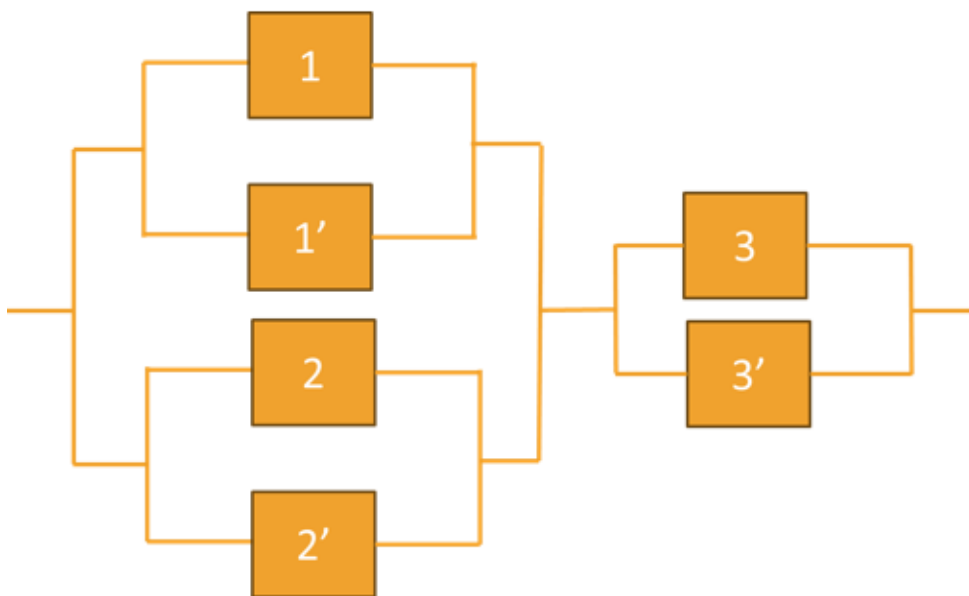
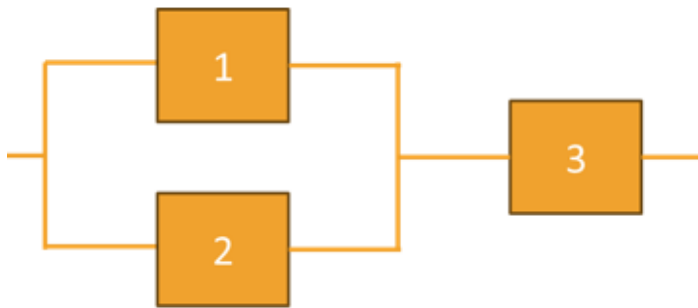
- ☐ a. 0.657
- ☐ b. 0.363
- ☒ c. 0.637
- ☐ d. 0.343

Question 26[Flag question](#)

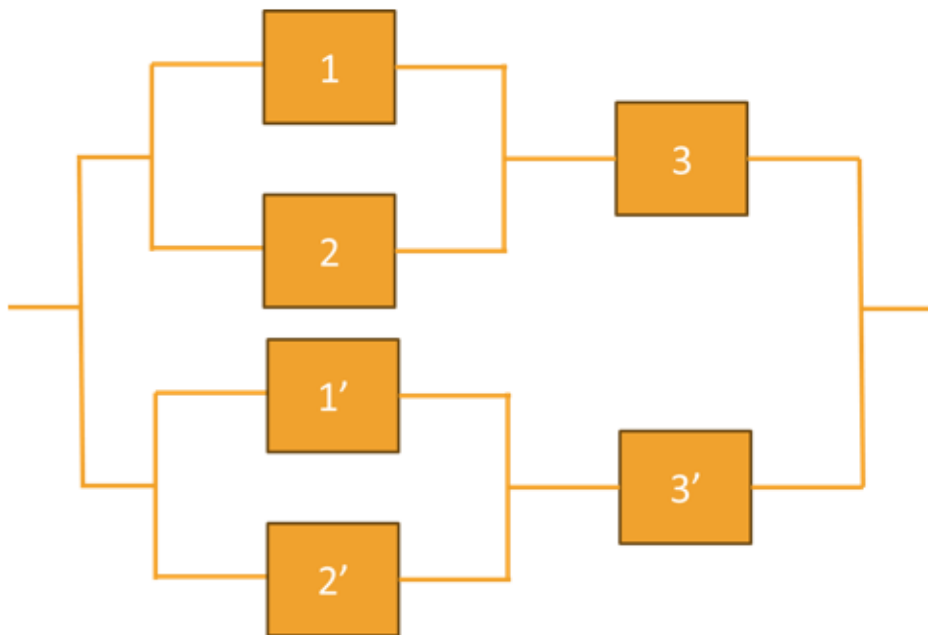
Complete

Marked out of 4.00

We have a system including 3 components as following figure. Assume that components have a single and identical failure mode. We need to enhance the system reliability by either using a component level redundancy as shown in figure (a) or a system level redundancy as shown in figure (b). Which structure is your recommendation for reliability enhancement?



(a) Component level redundancy



(b) System level redundancy

Select one:

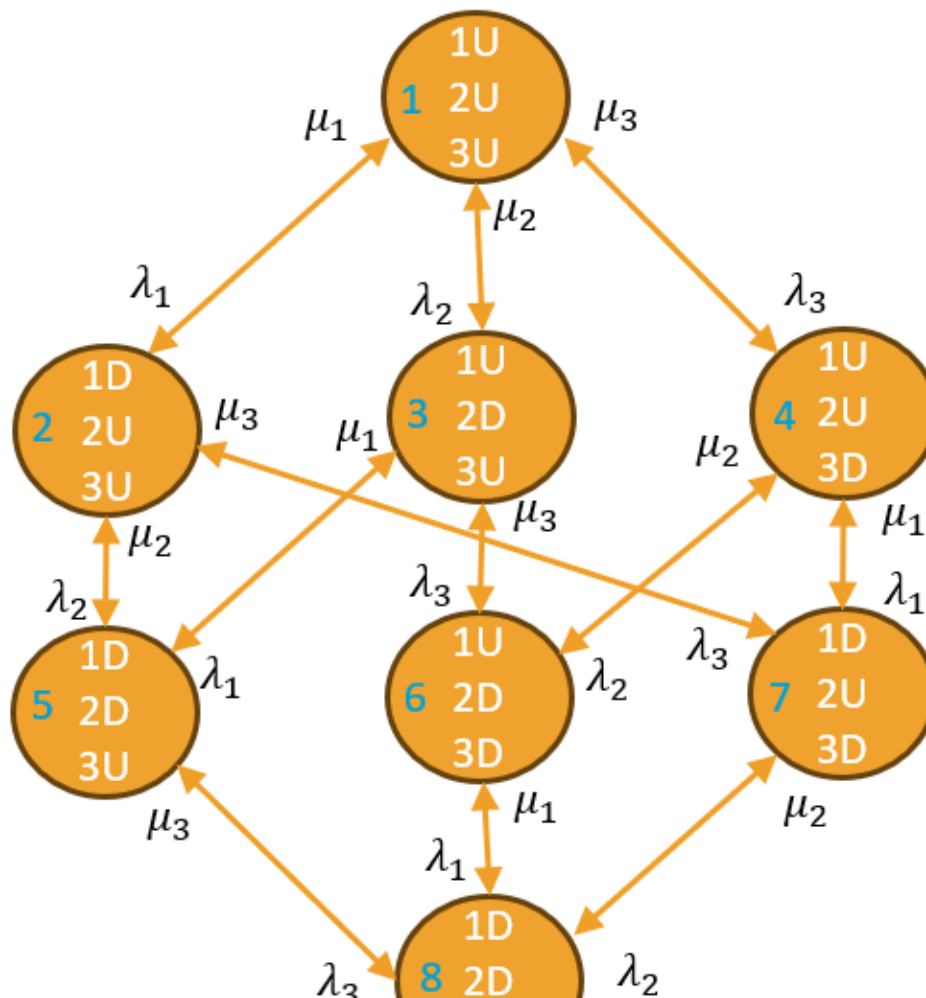
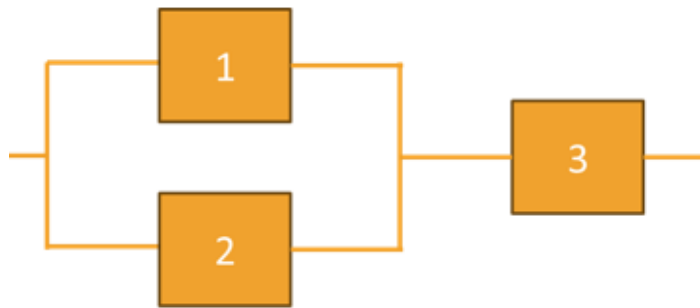
- ☐ a. Both will result in the same reliability.
- ☐ b. None of them
- ☒ c. Component level redundancy
- ☐ d. System level redundancy

Question 27[Flag question](#)

Complete

Marked out of 3.00

We have a system including 3 components as following figure. The Markov model is presented in the following figure, with the number of corresponding states mentioned in blue color on the figure. What are the successful states for this system?





Select one:

- ☐ a. 6, 7, 8
- ☒ b. 1, 2, 3
- ☐ c. 1, 2, 3, 4
- ☐ d. 1, 3, 6, 4
- ☐ e. 1, 3, 4

Question 28

[Flag question](#)

Complete

Marked out of 4.00

Assume that we have two generators with the capacity of 5 MW and 10 MW. The probability of each state is stated in the following table.

Generator 1 (5 MW)	Generator 2 (10 MW)	Probability
UP	UP	0.955
DOWN	UP	0.02
UP	DOWN	0.02
DOWN	DOWN	0.005

These generators will support a load of a factory that varies over a year. We simply model the factory load as 165 days 7.5 MW, 100 days 11 MW, and 100 days 4 MW.

Find the probability and expected duration that load level of 11 MW cannot be supplied due to unsuccessful states (meaning that due to failure or unavailability of generators).

Select one:

- ☐ a. Probability = 0.955
Expected duration = 100 days
- ☐ b. Probability = 0.045
Expected duration = 100 days
- ☒ c. Probability = 0.045
Expected duration = 4.5 days
- ☐ d. Probability = 0.955
Expected duration = 95.5 days

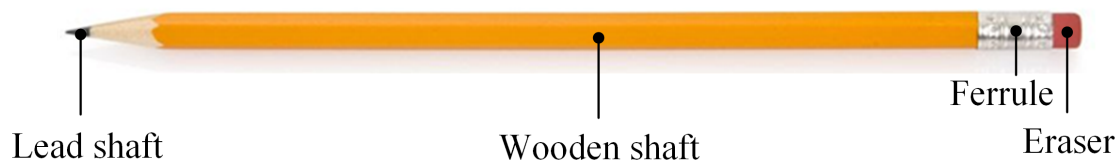
Question 29

[Flag question](#)

Complete

Marked out of 2.00

A team of members from a pencil company performs the design Failure Mode and Effect Analysis (FMEA) of a type of pencil shown in the figure below. The pencil has four parts: the lead shaft, wooden shaft, ferrule, and eraser.



If you are in charge of establishing the design FMEA team in the design phase of a new pencil product, who will you invite from the following categories of people (one or more choices):

Select one or more:

- ☒ a. Pencil manufacturing process control staff
- ☒ b. Project managers
- ☒ c. Pencil designers
- ☒ d. Testing engineers

Question 30

[Flag question](#)

Complete

Marked out of 2.00

In the following descriptions, which are the possible failure mode of the pencil (one or more choices)?

Select one or more:

- ☐ a. None of any of the other three descriptions
- ☒ b. User unable to write, with potential for minor injury from wood splinters
- ☒ c. Eraser falls off pencil during normal use
- ☒ d. Pencil lead breaks during normal usage

Question 31

[Flag question](#)

Complete

Marked out of 2.00

Severity, occurrence, and detectability are three parameters used for quantitative analysis in design FMEA. Which of the following do they describe for, respectively?

Select one:

- ☐ a. Failure mode, failure cause, failure cause
- ☒ b. Failure effect, failure cause, failure cause
- ☐ c. Failure effect, failure cause, failure effect
- ☐ d. Failure mode, failure cause, failure effect

Question 32

[Flag question](#)

Complete

Marked out of 2.00

Which of the following could be used as the recommended activities based on the design FMEA analysis (one or more choices)

Select one or more:



a. Perform Design of Experiment on eraser composition to optimize erasing performance.



b. Review pencil wood supplier process and ensure actions are in place to control the hardness of supplied wood to requirements.



c. Sharpened pencil lead extends too far from wood enclosure due to improper sharpening



d. Eraser falls off pencil during normal use

Question 33

[Flag question](#)

Complete

Marked out of 1.00

An interconnect component has thermal-mechanical fatigue mechanism. Its cycle-to-failure follows the following lifetime model defined at B10, where ΔT is the temperature variation amplitude.

$$N = 1.5 \times 10^{12} \times (\Delta T)^{-5}$$

The accumulated damage of 100 cycles of $\Delta T = 50^\circ\text{C}$ is (round to six decimal places, i.e., up to 0.000001):

Answer:

Question 34

[Flag question](#)

Complete

Marked out of 1.00

In connection with the previous question, the accumulated damage of 50 cycles of $\Delta T = 40^\circ\text{C}$ is (round to six decimal places, i.e., up to 0.000001):

Answer:

Question 35

[Flag question](#)

Complete

Marked out of 2.00

In connection with the previous two questions, assuming that each cycle last for 1 hour and the component operates with $\Delta T = 50\text{ }^{\circ}\text{C}$ for $2/3$ of the time, and with $\Delta T = 40\text{ }^{\circ}\text{C}$ for $1/3$ of the time, the B10 lifetime of the component is (round down to the nearest integer):

Answer:

6186

[Finish review](#)